

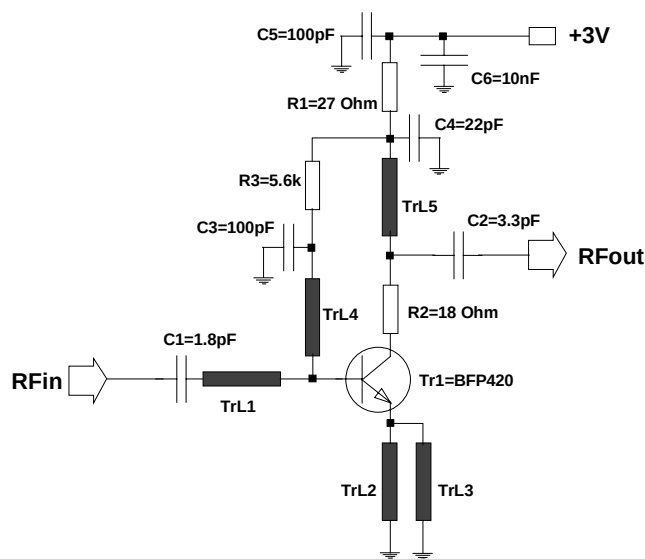
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A Low-Noise-Amplifier with good IP3out-performance at 1.9 GHz using BFP420

This application note describes a low noise amplifier at 1.9GHz using SIEMENS SIEGET[®] 25 BFP420. The design emphasis has been on achieving high output intercept point. General information, circuit schematic, pcb layout and components list are shown below together with measured performance data.

Data at 1.9 GHz (3V and 20mA):

Gain=15 dB / IP3out=23dBm / NF=1.9dB / R_{Lin-out} >15dB



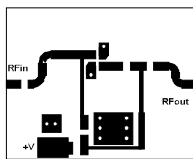
This amplifier at 1.9 GHz is realized by using microstrip lines as matching elements. The design offers a good compromise between high IP3-values, low noise figure- and high return loss values.

In order to optimise the design for a particular application please observe the following points:

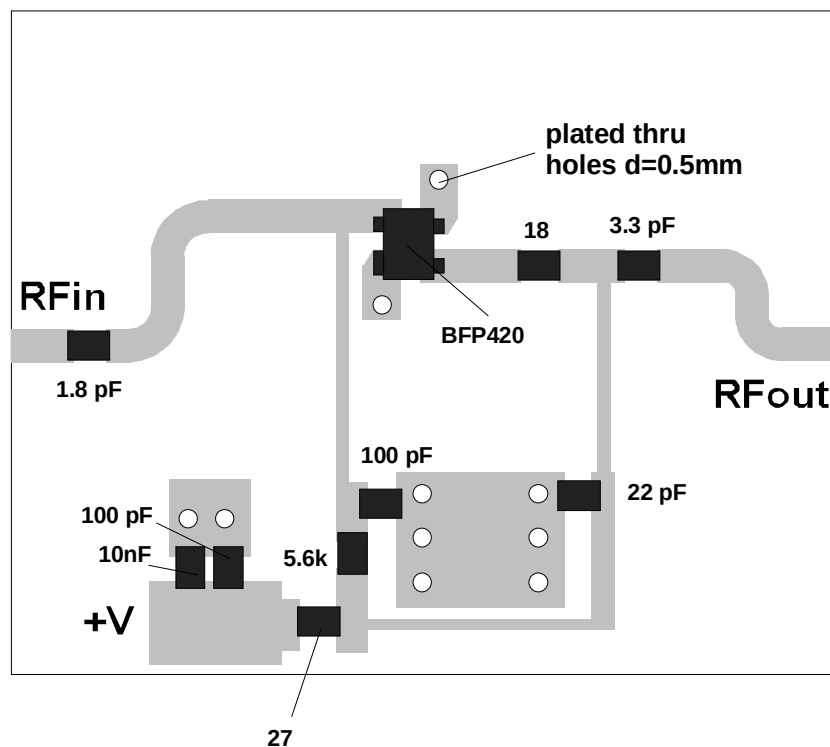
- The layout size can be reduced by using chip-inductors instead of the microstrip lines TrL1 and TrL4.
- Improved stabilization behaviour versus temperature and reduced variation in amplifier performance due to the device's Beta (current gain) distribution can be achieved by using an active bias circuit. Such a circuit is available as a single device from Siemens - BCR400. For further information please refer to Application Note 014. However, in most applications, sufficient stabilization is achievable using the resistors R1 and R2.
- This circuit is not optimized, it is only meant as a first step to a good design. The measured figures include losses of SMA-connectors and the relatively high loss of the microstrip lines on the epoxy-board.
- The use of teflon material would provide an improvement of ≈ 0.1 dB.
- Resistor R2 is used to improve rf-circuit-stability and return loss values at the output. It also affects the output intermodulation performance.

Component	Value	Unit	Size	Comment
R1	27	Ω	0603	bias / collector-resistance / $VR1 \cong 0.5V$
R2	18	Ω	0603	to improve stability and output return loss
R3	5.6	k Ω	0603	bias / base-resistor
C1	1.8	pF	0603	inputmatch
C2	3.3	pF	0603	outputmatch
C3	100	pF	0603	rf-short
C4	22	pF	0603	outputmatch
C5	100	pF	0603	rf-short
C6	10	nF	0603	rf-short
Tr1			SOT343	SIEGET BFP420
TrL1				inputmatch , $w=0.95mm$
TrL2				emitter-microstrip-line, $w=0.95mm$
TrL3				emitter-microstrip-line, $w=0.95mm$
TrL4				inputmatch and dc-bias, $w=0.3mm$
TrL5				outputmatch and dc-bias, $w=0.3mm$
Substrate	FR4			$h=0.5mm$; $\epsilon_r=4.5$

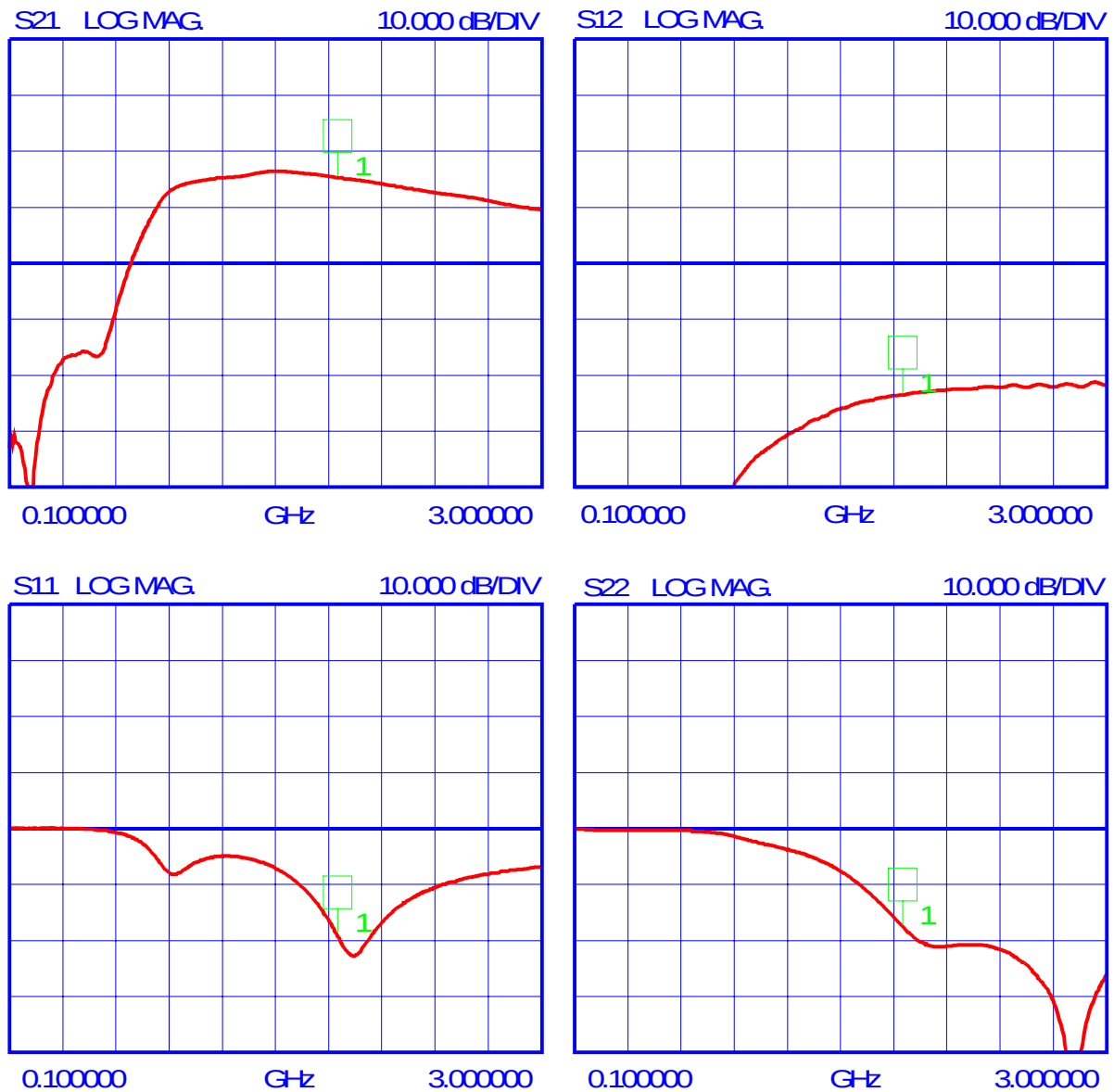
PCB Layout and Component Placement



scale 1:1
dim.: 25mm x 20mm



Measured data :



1 1.9 GHz

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